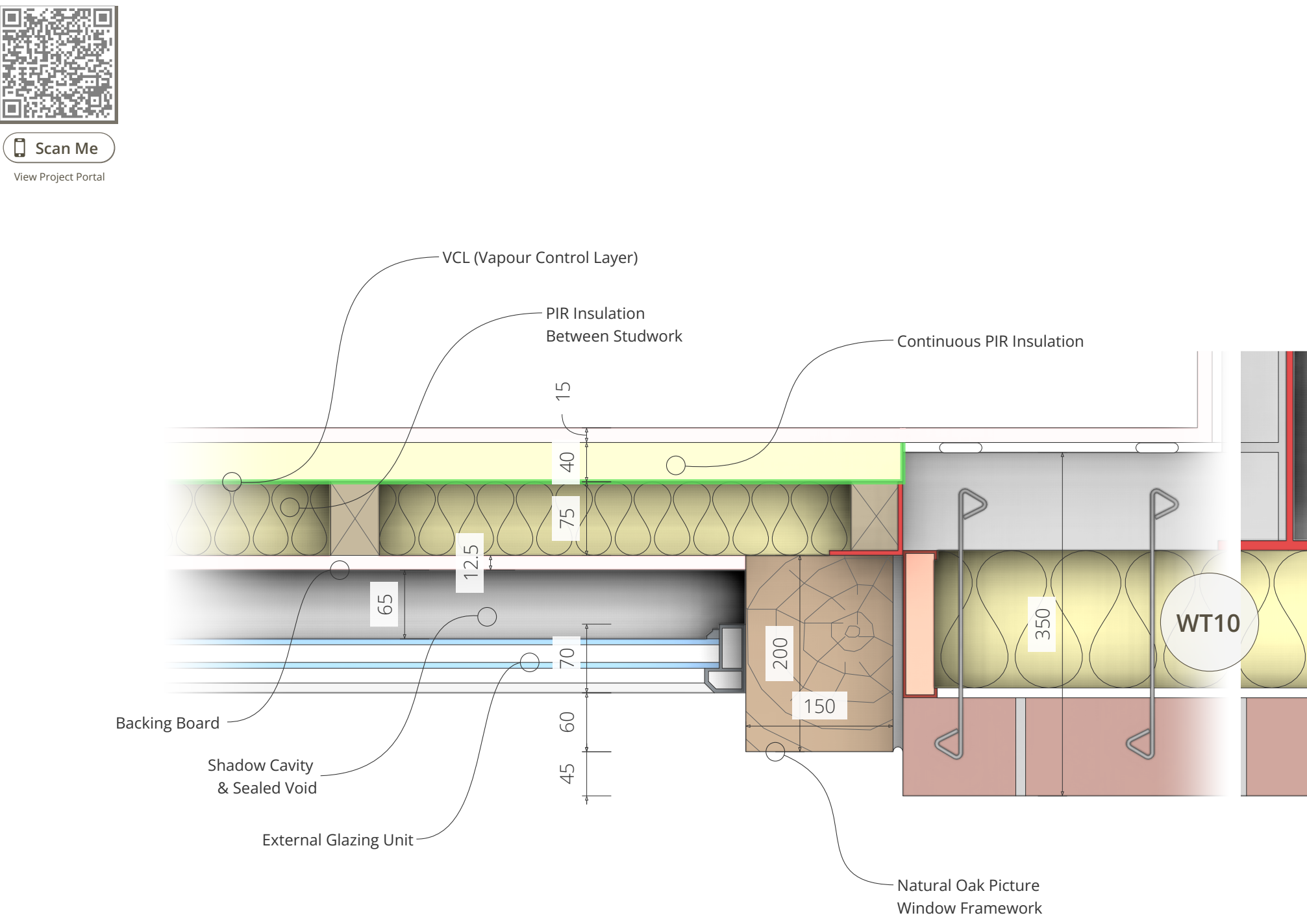




WT30 - Insulated Shadow Box Spandrel Detail

Calculation			
U = 1 / (0.04 + 0.83 + 0.18 + 0.07 + 2.32 + 1.82 + 0.07 + 0.13) = 1 / 5.45 = 0.18 W/m²K			
Layer	Thickness (mm)	Lambda (W/mK)	R-value (m²K/W)
External Surface Resistance (Rse)	-	-	0.040
External Glazing Unit (Double Glazed - Heat Strengthened)	28	-	0.833
Shadow Cavity - Sealed Air Space (Unventilated)	65	-	0.180
Backing Board - 12.5mm MR Plasterboard	12.5	0.190	0.066
Insulation - Primary PIR Between Studs (Bridged 10%)	75	0.028	2.320
Insulation - Secondary Continuous PIR (Unbridged)	40	0.022	1.818
VCL - Foil Face of PIR (Joints Taped)	-	-	0.000
Internal Lining - Plasterboard & Skim	15.5	0.190	0.066
Internal Surface Resistance (Rsi)	-	-	0.130
Total R-value			5.453
Result			
U-Value			0.18 W/m²K

Value Compliant - England Part L1B : Wall U-value 0.18 W/m²K.

Element	Specifications
System Description	High-Performance Insulated 'Shadow Box' Spandrel Panel. A hybrid construction combining an external obscured glazed rain-screen with a thermally insulated internal timber-frame backing wall. Designed to provide a 'Continuous Solid Wall' appearance internally (skirting/plaster) while matching the external glazing rhythm.
External Glazing Unit	Outer Pane: 6mm Heat Strengthened Clear Float. Cavity: 16mm Argon Filled (90%) with Warm Edge Spacer (Black). Inner Pane: 6mm Heat Strengthened OBSCURED Float (Acid Etched/Satin finish on Surface #3 or #4). Coating: High-Performance Low-E coating on Surface #2. Rationale: The South-facing orientation and large pane size will generate rapid solar gain and extreme thermal cycling. 'Heat Strengthened' glass is mandatory to resist the resulting thermal stress and high stagnation temperatures (>80°C) which would shatter standard annealed glass.
Shadow Cavity	65mm nominal sealed air space. STRICTLY NO VENTILATION to exterior or interior. This depth creates the optical 'Parallax Effect', giving the panel visual depth rather than a flat appearance.
Moisture Control	Install Molecular Sieve Desiccant sachets (min. 50g/m²) secured in the bottom rebate of the cavity frame. Function: To scavenge any residual moisture or volatiles trapped during assembly, preventing 'fogging' or condensation on the inner glass face.
Backing Board	12.5mm Moisture Resistant Plasterboard (Green Face). Finish: Taped and Jointed to Level 5 standard (flawless). Decoration: 2 coats Zinsser Perma-White (Satin) or equivalent Exterior Grade heat/mould resistant paint. Rationale: Replaces MDF to eliminate the risk of Urea-Formaldehyde off-gassing (chemical fogging) in the heated cavity and improves the fire classification of the backing layer.
Primary Insulation	75mm Celotex GA4000 (General Application PIR). Friction fit tightly between the timber studwork with no gaps.
Internal Structure	75mm x 47mm C16 Timber Studs at 600mm centres. The 600mm spacing is specified to reduce the volume of timber bridging compared to standard 400mm centres.
Secondary Insulation	40mm Celotex GA4000 fixed continuously across the internal face of the studs. This continuous layer is critical to break the thermal bridge of the timber studs, ensuring the assembly performs as a thermally broken wall rather than a cold frame.
Internal Lining	12.5mm Plasterboard + 3mm Thistle Multi-Finish Skim. Finished with skirting board to match the room.
Design Rationale	1) The 'Dummy Room' Effect: The 65mm cavity, combined with obscured glass and a white backing, creates a luminous volume that diffuses daylight. This mimics the depth and light-play of an occupied room, preventing the 'dead/black' appearance typical of flush spandrel panels. 2) Balanced Thermal Design: By utilising 75mm PIR between studs and a 40mm continuous layer over them (Total 115mm PIR), the design maximizes U-value performance within the restricted depth, without compromising the critical 65mm shadow cavity required for the aesthetic effect. 3) Material Safety: The switch to Moisture Resistant Gypsum backing removes the primary failure mode of shadow boxes (MDF chemical fogging) and enhances fire safety.
Ventilation Strategy	STRICTLY NO VENTILATION. Do not install grilles in the window board or external frame. 1) Internal venting creates a 'Condensation Trap' where warm, moist internal air strikes the cold outer glass. 2) External venting allows dust/insect ingress. 3) Any vent creates a thermal bypass. The unit must be a fully SEALED system relying on the desiccant and sealed backing to prevent fogging.
Part L Classification	Regulatory Definition: Opaque Wall Element (NOT Controlled Fitting/Window).Justification: Despite the external glass rain-screen finish, this assembly functions physically and thermally as an opaque timber-frame wall.1) Light Transmission: 0% (Opaque backing).2) Thermal Performance: U-Value 0.18 W/m²K (Matches Part L 'Wall' threshold, far exceeding the 1.2 W/m²K 'Window' threshold).Calculation Status: For the purposes of SAP and the '25% Glazing Rule' (Approved Document L, Vol 1), this area is EXCLUDED from the total glazing area calculation. It is strictly defined as an insulated wall with a glass rain-screen cladding.

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.